

DBMS.

unit 1

Assignment 1.

Q1.) List and explain four characteristics of DBMS. Details

Four Characteristics of DBMS (Detailed Explanation)

1 Data Independence

Definition:

Data independence means the database structure can be changed without affecting application programs.

Explanation:

In traditional file systems, if the structure of data changes, programs must also change. In DBMS, data is separated from application programs. This separation allows modification in database schema without rewriting software.

Types of Data Independence

- **Physical Data Independence** → Changes in storage (file organization, indexing) do not affect users.
- **Logical Data Independence** → Changes in table structure do not affect application programs.

Example:

If a new column “Email” is added in Student table, existing programs still work.

Importance:

- ✓ Reduces program maintenance
- ✓ Provides flexibility
- ✓ Supports system growth

2 □ Data Integrity

Definition: Data integrity ensures accuracy, consistency, and correctness of data stored in the database.

Explanation: DBMS applies rules called constraints to maintain valid data. It prevents wrong or duplicate data from entering the database.

Types of Integrity Constraints

Entity Integrity → Primary key cannot be NULL

Referential Integrity → Foreign key must match parent table

Domain Integrity → Values must be in valid range

Example:

Marks cannot be negative.

Student ID must be unique.

Importance:

✓ Reliable information

✓ Consistent database

✓ Error prevention

3 Data Security

Definition:

Data security protects database from unauthorized access and misuse.

Explanation:

DBMS provides authentication and authorization mechanisms. Only authorized users can view or modify data.

Security Methods

- User login and password
- Access privileges (read, write, update)
- Encryption
- Backup protection

Example:

Only admin can delete records; students can only view results.

Importance:

- ✓ Protects confidential data
- ✓ Prevents data theft
- ✓ Ensures privacy

4. Control of Data Redundancy

Definition:

Data redundancy means unnecessary duplication of data. DBMS minimizes this duplication.

Explanation:

Instead of storing same data in multiple files, DBMS stores it once and links tables using keys. This avoids inconsistency.

Example:

Student details stored in one table and referenced by roll number in other tables.

Problems Without Control

- ✗ Data inconsistency
- ✗ Waste of storage
- ✗ Update anomalies

Importance:

- ✓ Saves storage space
- ✓ Improves consistency

Q2) Explain Database users. in Details

Database users are people who interact with the database system in different ways depending on their roles and responsibilities.

1 Database Administrator (DBA)

Definition:

The DBA is the person who has full control over the database system.

Responsibilities:

- Defines database schema and structure
- Controls user access and permissions
- Ensures data security and integrity
- Performs backup and recovery
- Monitors performance and storage
- Handles database failures

Importance:

DBA ensures the database runs smoothly, securely, and efficiently.

Example:

In a college system, DBA manages student records database and gives access to staff

2 Application Programmers (Developers)

Definition:

Users who write application programs that interact with the database.

Responsibilities:

- Develop software using database queries
- Create forms, reports, and interfaces
- Implement business logic
- Use SQL inside programs

Example:

A programmer develops a result management system connected to the database.

Importance:

They act as a bridge between end users and the database

3 End Users

End users are people who access the database through applications or interfaces.

Types of End Users



Naive (Parametric) Users

- Use predefined forms
- Do not know database structure
- Perform routine operations

Example:

ATM users, students checking results

4. System Analysts (Sometimes Included)

Definition:

Professionals who design how the database system should work.

Responsibilities:

- Analyze user requirements
- Design database structure
- Coordinate between users and developers

Q3)What is DBMS?

DBMS (Database Management System) is software that allows users to create, store, organize, retrieve, and manage data in a structured way.

It acts as an interface between:

- ☐ Database (data storage)
- ☐ Users / Applications

Examples: MySQL, Oracle, SQL Server, PostgreSQL

Functions of DBMS:

Data storage and retrieval

Data security and authorization

Backup and recovery

Data integrity control

Advantages of DBMS Approach

✓1. Data Redundancy Control

Duplicate data is minimized → saves storage and avoids inconsistency.

✓2. Data Consistency

Same data is shared across applications → updates reflect everywhere.

✓3. Data Security

Access permissions and authentication protect sensitive data.

✓4. Data Integrity

Rules and constraints ensure accurate and valid data.

✓5. Data Sharing

Multiple users can access data simultaneously.

✓6. Backup and Recovery

Automatic recovery after system failure.

✓7. Data Independence

Changes in database structure do not affect

Disadvantages of DBMS Approach

✗1. High Cost

Software, hardware, and maintenance are expensive.

✗2. Complexity

Designing and managing DBMS requires skilled professionals.

✗3. Large Size

DBMS requires more memory and storage.

✕4. Performance Overhead

For small applications, DBMS may be slower than simple file systems.

✕5. Single Point Failure Risk

If DBMS fails, whole system access may stop.

Q4).Difference: Database System vs Conventional File System

Basis	Database System (DBMS)	Conventional File System
-----	-----	-----
Data Redundancy	Controlled / minimized	High redundancy
Data Consistency	High consistency	Poor consistency
Data Security	Strong security control	Limited security
Data Sharing	Multi-user access supported	Difficult to share
Data Independence	Supported	Not supported
Backup & Recovery	Automatic recovery system	Manual backup
Data Integrity	Constraints maintained	No proper control
Program-Data Relation	Independent	Highly dependent
Complexity	More complex	Simple
Cost	High	Low

Q5). Explain Hierarchical Model in Detail

□ Definition. Hierarchical Data Model ek tree structure par based database model hai jisme data parent → child relationship me organize hota hai. Har child ka sirf ek parent hota hai, lekin parent ke multiple children ho sakte hain.

□ Structure

Top record → Root

Parent record → Upper level

Child record → Lower level

One-to-many relationship

Example structure:

Company

├── Department

| ├── Employee

| ├── Employee

├── Department

├── Employee

□ Characteristics

Tree-like structure

One parent → many children

Fast data access through path

Predefined relationship

No many-to-many relation

☐ Advantages

✓ Simple structure

✓ Fast retrieval

✓ Data security high

✓ Logical parent-child control

☐ Disadvantages

✗ Rigid structure

✗ Complex implementation

✗ No many-to-many relation

Q6). Explain Network Model & Relational Model

☐ Network Data Model

Definition

Network model me data records graph structure me connected hote hai. Ek child ke multiple parents ho sakte hai.

Structure

Records = Nodes

Relationships = Links

Many-to-many relationship supported

Example:

Student $\longleftrightarrow$ Course

Ek student multiple course le sakta hai, ek course multiple student ka ho sakta hai.

Characteristics

Graph structure

Many-to-many relation

Flexible structure

Pointer based connection

Advantages

✓ More flexible than hierarchical

✓ Data redundancy kam

✓ Efficient relationship handling

Disadvantages

✗ Complex design

✗ Navigation difficult

Relational Data Model

Definition

Relational model me data tables (relations) me store hota hai. Rows = tuples, Columns = attributes.

Structure

Example table:

StudentID	Name	Course
101	Amit	DBMS

Key Concepts

Primary Key → Unique identifier

Foreign Key → Relationship between tables

Relation → Table

Tuple → Row

Attribute → Column

Characteristics

Table format storage

Data independence

Easy querying using SQL

Logical relationship using keys

Advantages

- ✓ Simple structure
- ✓ Easy data manipulation
- ✓ Flexible queries
- ✓ No navigation problem

Disadvantages

- ✗ Complex join operations

Q7 Explain E-R Model in Detail

☐ Definition. Entity-Relationship Model database design ka conceptual model hai jo real world objects aur unke relationships ko represent karta hai.

☐ Components

1 ☐ Entity

Real world object

Example: Student, Course

2 □ Attribute

Entity ki properties

Example: Student → Name, Roll No

Types:

Simple attribute

Composite attribute

Derived attribute

Multivalued attribute

3 □ Relationship

Entities ke beech connection

Example: Student enrolls Course

4 □ Cardinality

One-to-One

One-to-Many

Many-to-Many

□ ER Diagram Symbols

Rectangle → Entity

Oval → Attribute

Diamond → Relationship

Line → Connection

☐ Advantages

✓ Easy to understand

✓ Graphical representation

✓ Database design me useful

☐ Disadvantages

✗ No direct implementation

Q8). Explain DBMS Languages (DDL, DML, DCL)

☐ 1. DDL — Data Definition Language

Purpose

Database structure define ya modify karna.

Commands

CREATE → table create

ALTER → table modify

DROP → table delete

TRUNCATE → data remove

Example

```
CREATE TABLE Student(  
  ID INT,  
  Name VARCHAR(20)  
);
```

Features

- ☐ Structure define karta hai
- ☐ Auto commit hota hai

2. DML — Data Manipulation Language

Purpose

Table ke andar data manage karna.

Commands

SELECT → data retrieve

INSERT → data add

UPDATE → data change

DELETE → data remove

Example

```
INSERT INTO Student VALUES(101,'Amit');
```

```
SELECT * FROM Student;
```

Features

- ☐ Data handling
- ☐ Transaction control possible

3. DCL — Data Control Language

Purpose

Database security aur permissions control karna.

Commands

GRANT → permission dena

REVOKE → permission remove

Example

```
GRANT SELECT ON Student TO user1;
```

```
REVOKE SELECT ON Student FROM user1;
```

Features

- ☐ Access control

Q9)Centralized Architecture of DBMS

Meaning:

In a centralized database system, the entire database and DBMS software are stored and processed at one central server. All users access it through terminals or client machines.

Working:

One powerful computer = stores database + runs DBMS.

Users connect via network/terminals.

Processing, security, backup → all handled at central site.

Characteristics:

Single location data storage

Central control of operations

Simple management

All requests processed by one system

Advantages:

- ☐ Easy data management
- ☐ Strong security control
- ☐ Low maintenance cost
- ☐ Data consistency is high

Disadvantages:

- ☐ Single point of failure
- ☐ Performance decreases with many users

Q10)Client–Server Architecture of DBMS

Meaning:

Database is stored on a server, and multiple clients (user computers) send requests to the server to access data.

Working:

Client → sends request (query)

Server → processes request

Server → sends result back to client

Types:

- ☐ Two-Tier Architecture

Client directly communicates with database server.

Flow:

Client Application → Database Server

Use: Small organizations, LAN based systems

Classification of DBMS

DBMS are classified based on data model.

Main Types:

Hierarchical DBMS

Network DBMS

Relational DBMS

Object-Oriented DBMS

Explain Any Two Types of DBMS

☐ Hierarchical DBMS

Structure: Tree-like structure (Parent → Child relationship)

Features:

One parent can have many children

Child has only one parent

Data stored in levels

Advantages:

- ☐ Fast data retrieval
- ☐ Simple structure

Disadvantages:

- ☐ No many-to-many relationship

Relational DBMS (Most Important for Exam ☐)

Structure: Data stored in tables (relations).

Components:

Rows → Records (Tuples)

Columns → Attributes

Table → Relation

Features:

Uses SQL language

Supports relationships between tables

High data independence

Advantages:

- ☐ Easy to use
- ☐ Flexible data access
- ☐ Data integrity maintained

Disadvantages:

- ☐ Complex design for very large data